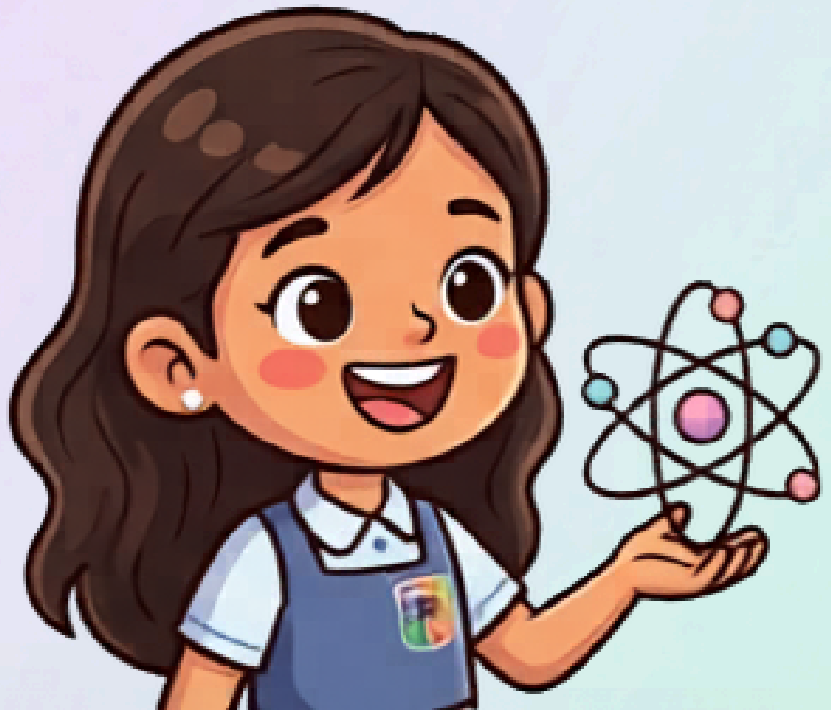
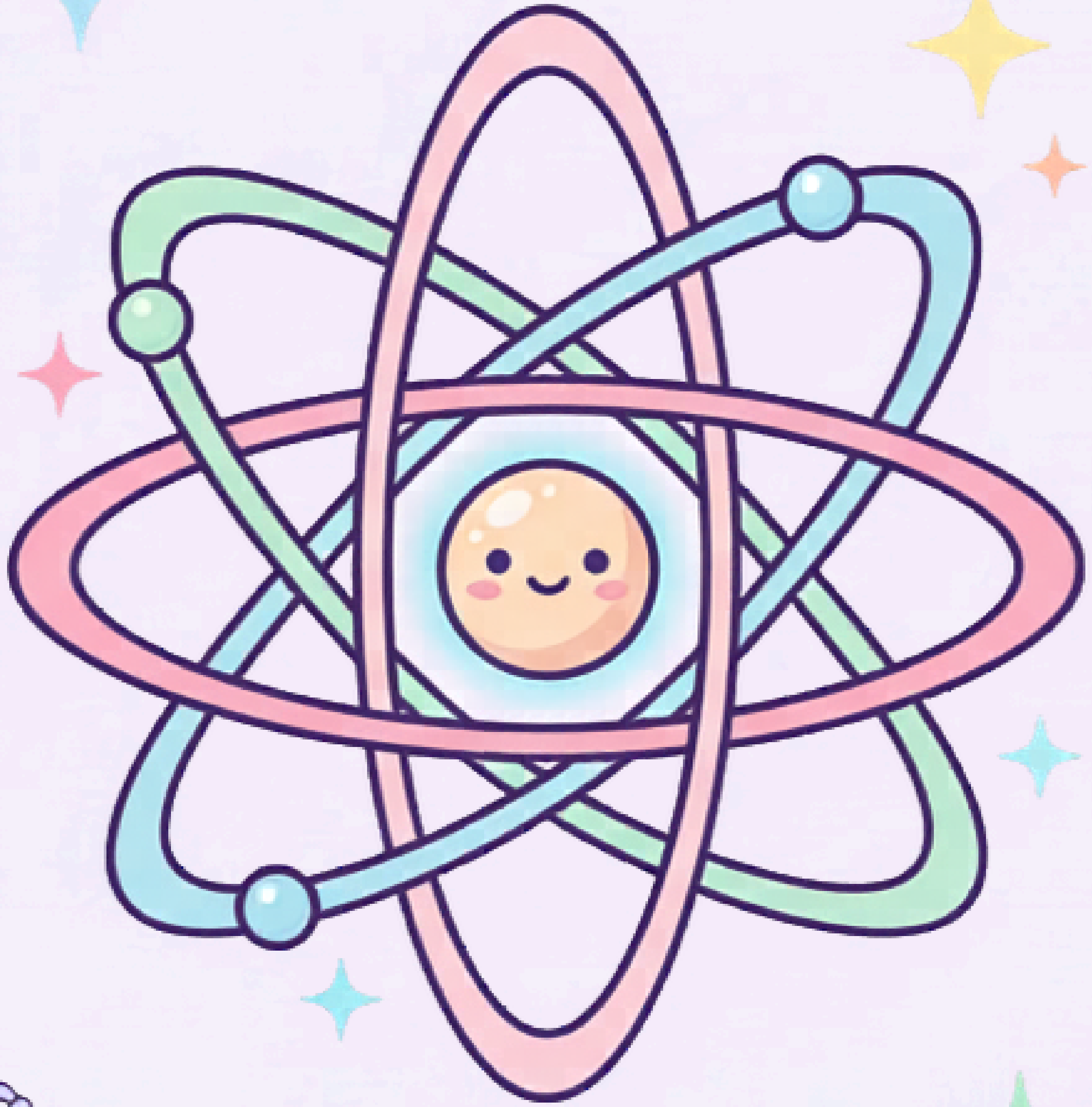
NUCLEAR ENERGY: FISSION & FUSION

Grade 6 STEM | Engaging Science for You



What is Nuclear Energy?



Nuclear energy is a special kind of power that comes from the very center of atoms, called the nucleus! It's one of the most powerful energy sources we know. Scientists use this amazing energy to make electricity that powers our homes, schools, and cities. The energy is released when changes happen inside the tiny nucleus of an atom.

Atoms – The Building Blocks

Atoms are tiny particles that make up everything around us – from the air we breathe to the food we eat!

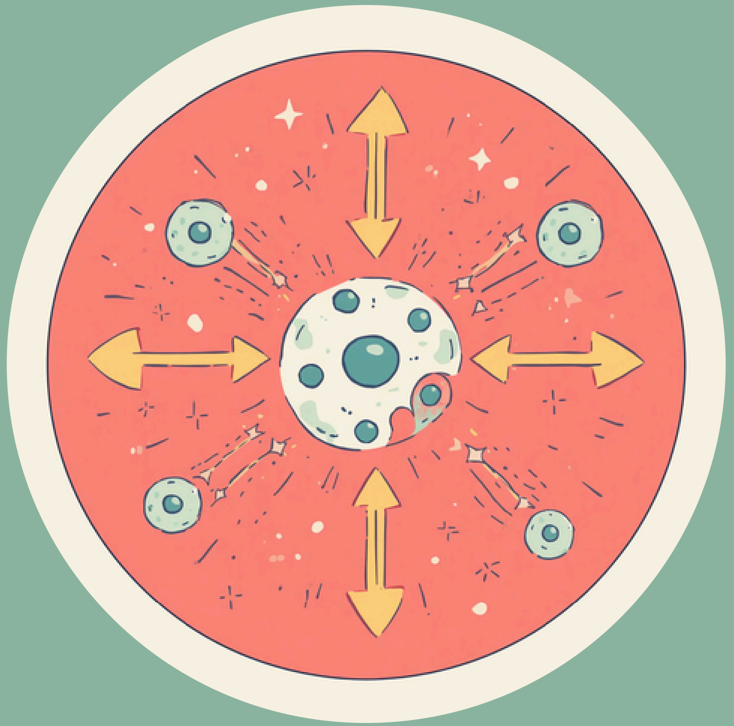
Every atom has a center called the nucleus. Inside the nucleus, you'll find protons and neutrons packed together.

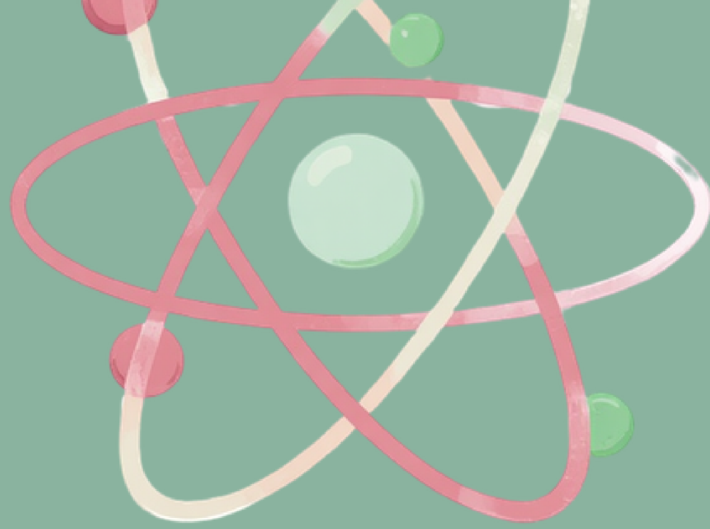
Electrons zoom around the nucleus in a cloud.

These building blocks are so small, millions could fit on the tip of a pencil!

What is Nuclear Fission?

Fission means splitting a big nucleus into smaller parts. Imagine a large, heavy atom like Uranium. When this atom's nucleus is hit by a tiny particle called a neutron, something amazing happens – the nucleus breaks apart into smaller pieces!
When the nucleus splits, it releases a huge amount of energy. This energy can be used to make electricity and power many things we use every day. It's like breaking open a piñata – when it splits, all the goodies come out!

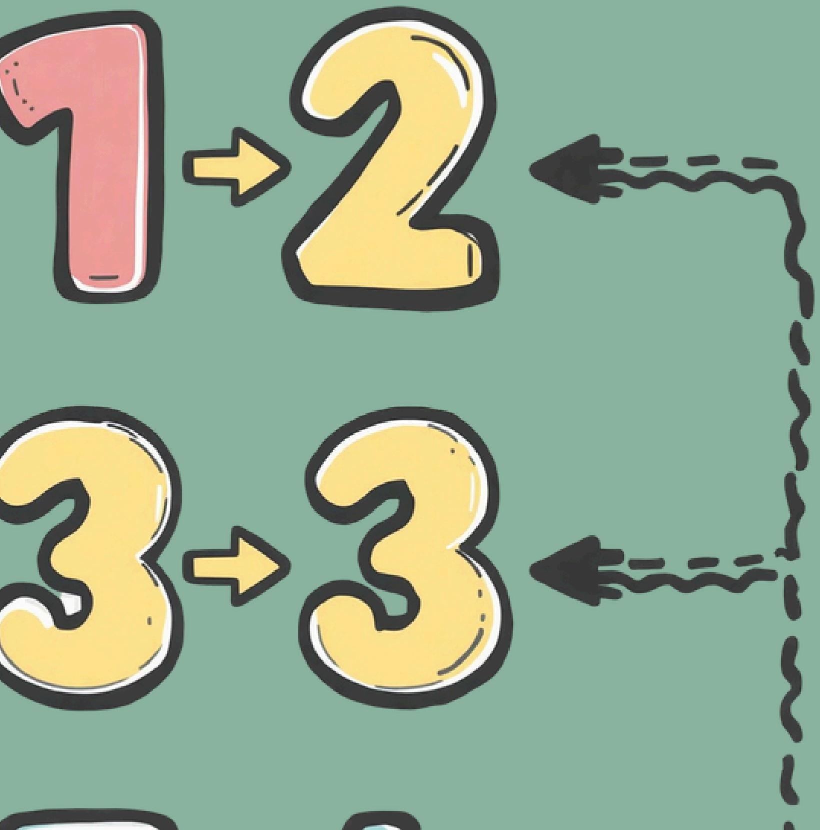


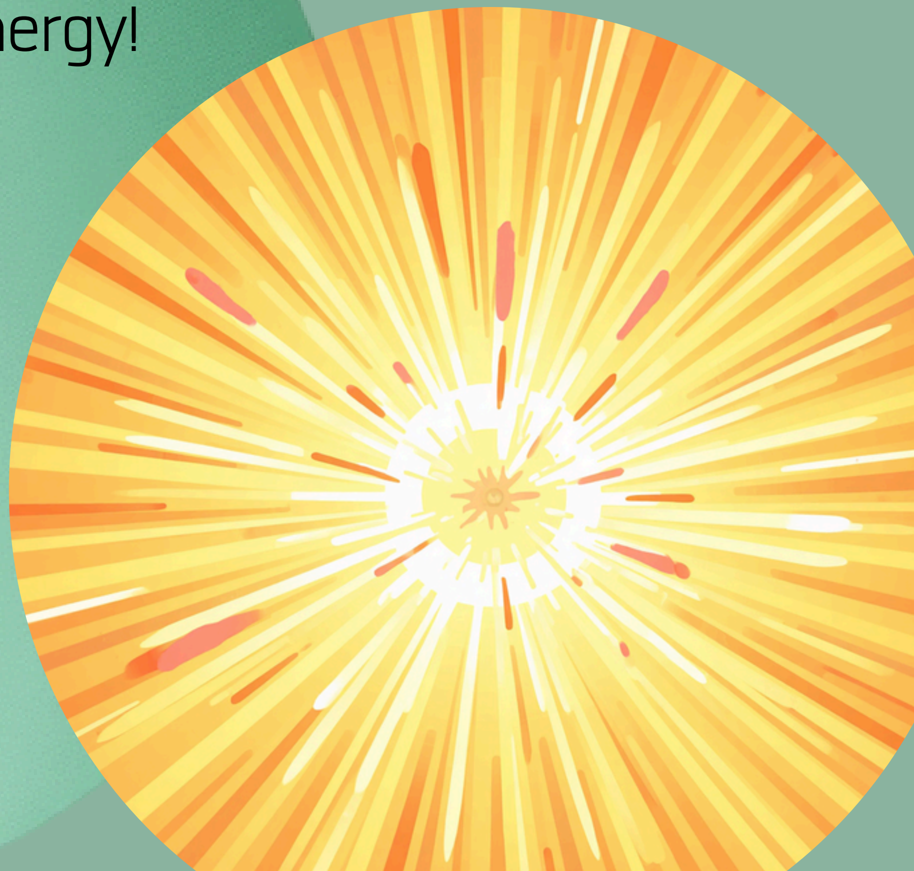


How Does Fission Work?

1. A neutron hits a big nucleus (like Uranium)
2. The nucleus splits into smaller nuclei
3. Energy and more neutrons are released
4. The released neutrons hit other nuclei, causing a chain reaction

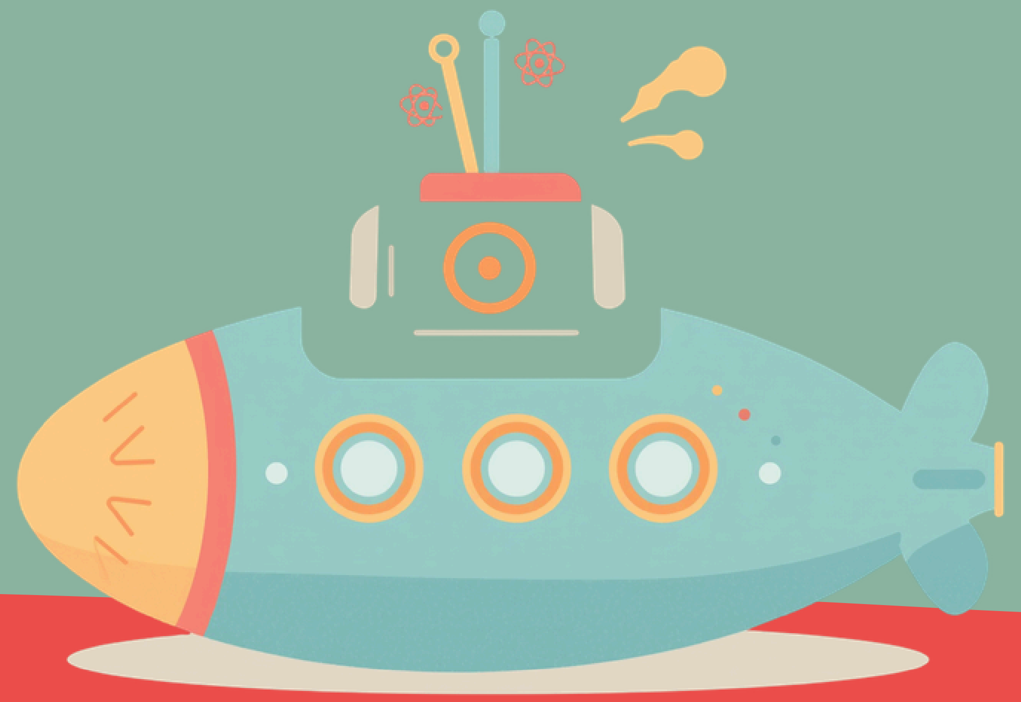
This chain reaction keeps going, releasing more and more energy!

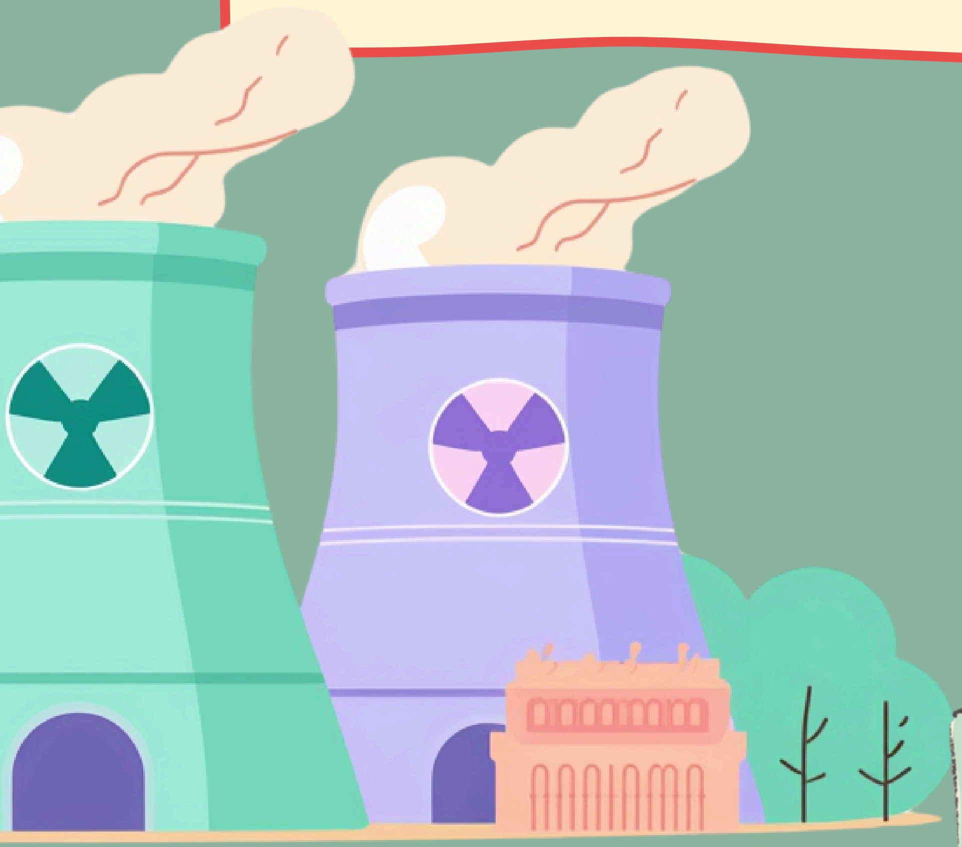


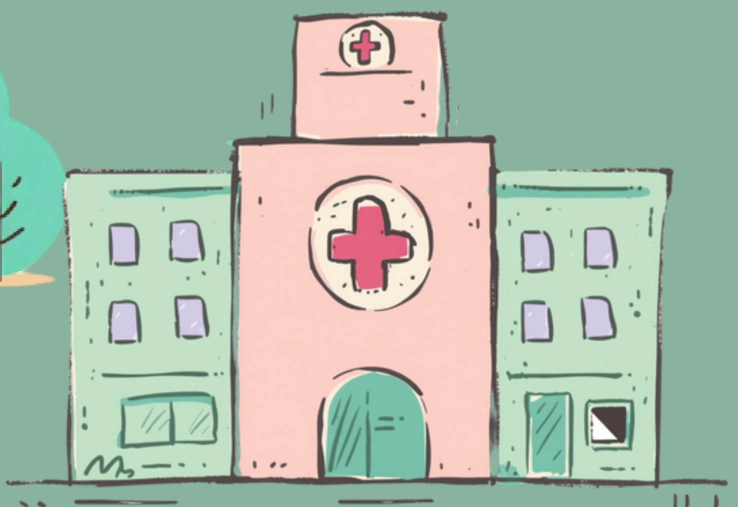


Uses of Nuclear Fission

Nuclear fission is used in many important ways! Power plants use fission to make electricity for millions of homes. The energy from splitting atoms heats water to create steam, which spins turbines to generate power. It's one of the most powerful ways we make electricity today!



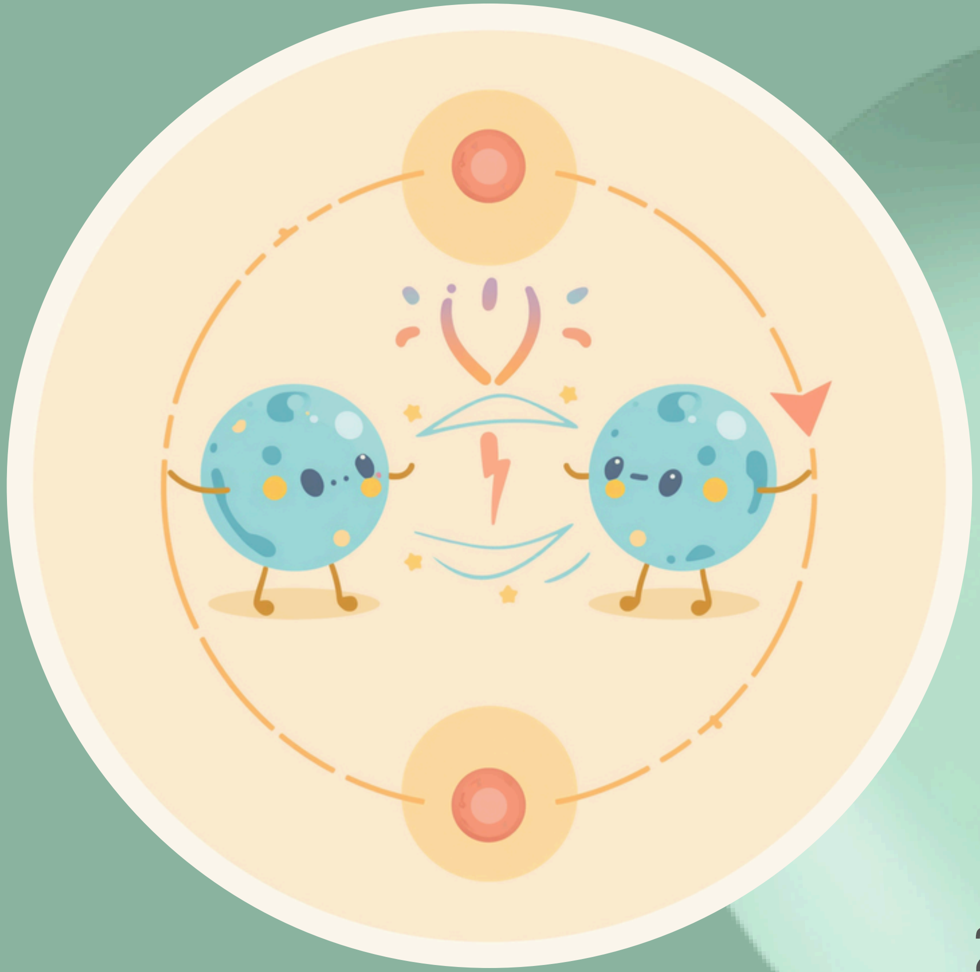


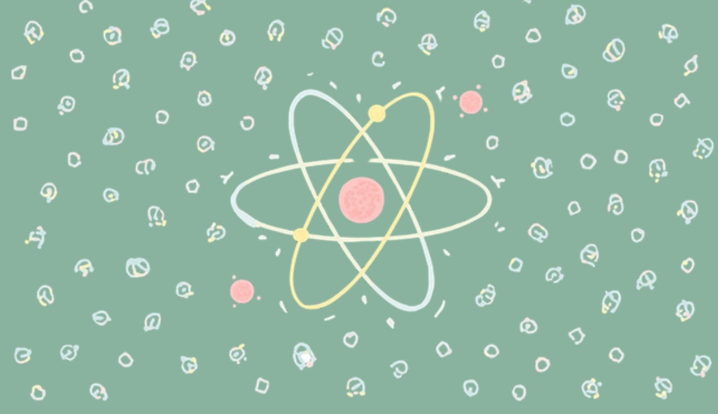


Fission also helps in medicine! Doctors use nuclear energy to treat cancer and take special pictures inside our bodies. Ships and submarines can travel for years without refueling because they use nuclear power. Fission energy is truly amazing!

What is Nuclear Fusion?

Fusion means joining small nuclei together to make a bigger nucleus. Imagine two tiny hydrogen atoms coming together and combining into one! When this happens, a huge amount of energy is released – even more energy than nuclear fission! This is the same process that powers our Sun and all the stars in the universe. Scientists are working hard to use fusion energy here on Earth because it could give us clean, powerful energy for the future.

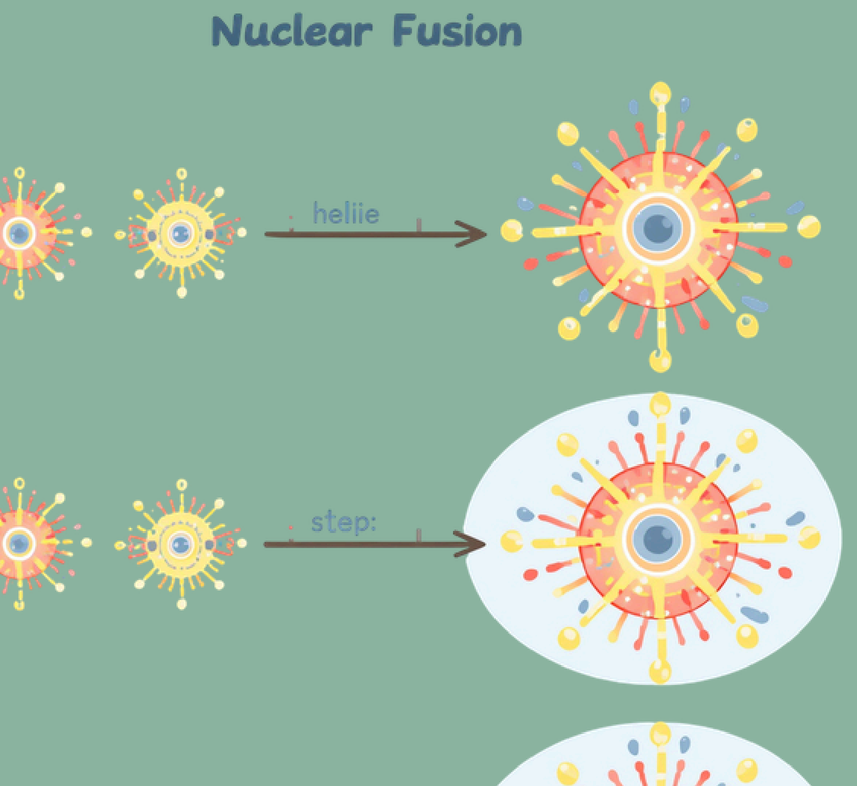


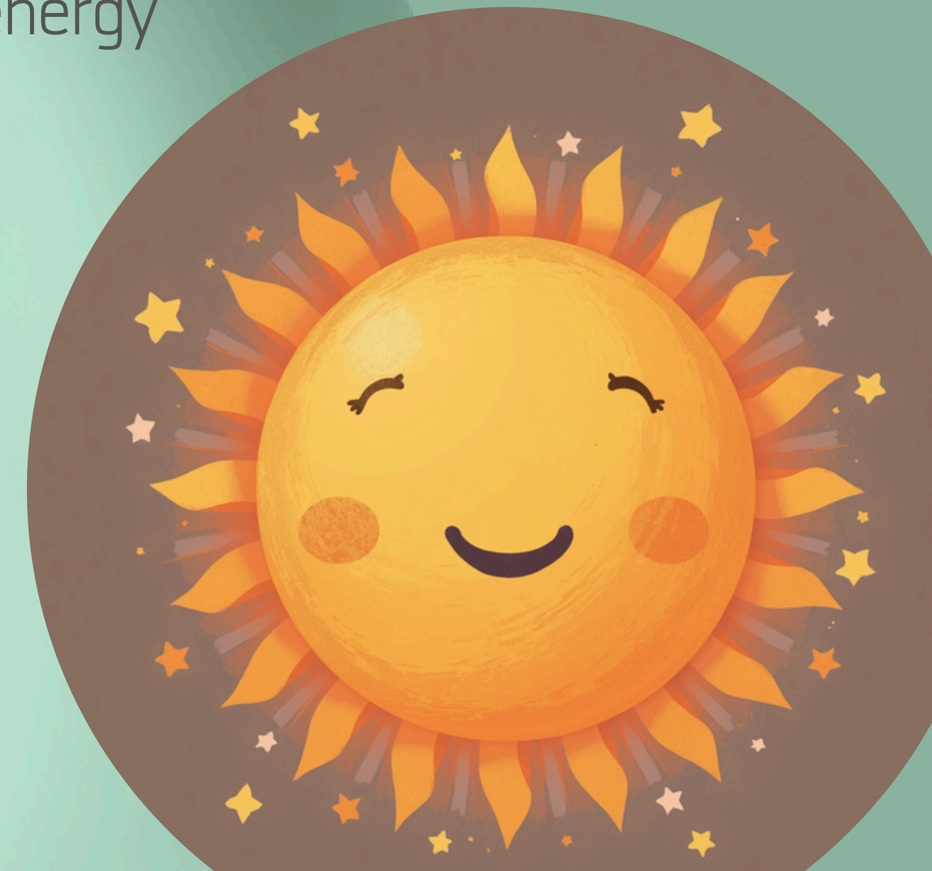


How Does Fusion Work?

1. Two small nuclei, like hydrogen atoms, come very close together
2. They join to form a bigger nucleus (like helium)
3. A huge amount of energy is released!

Fusion powers the Sun and all the stars in the sky. The Sun creates energy through fusion every single second, giving us light and warmth!





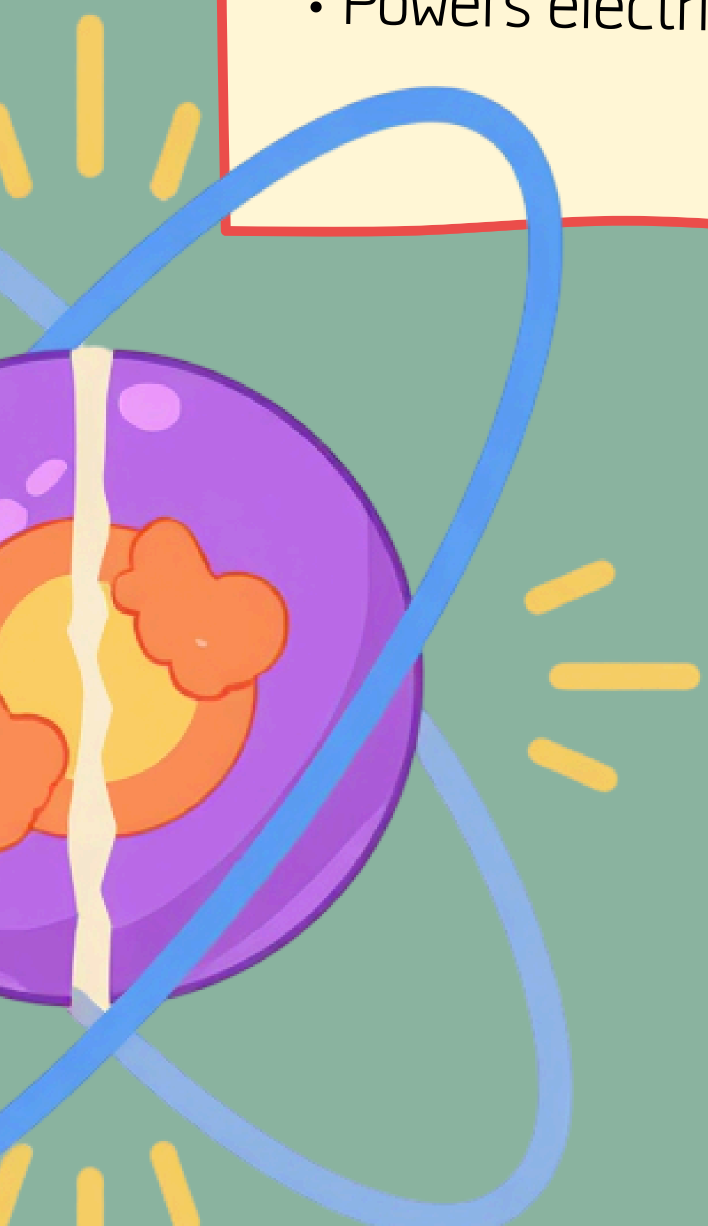
Uses & Potential of Fusion Energy

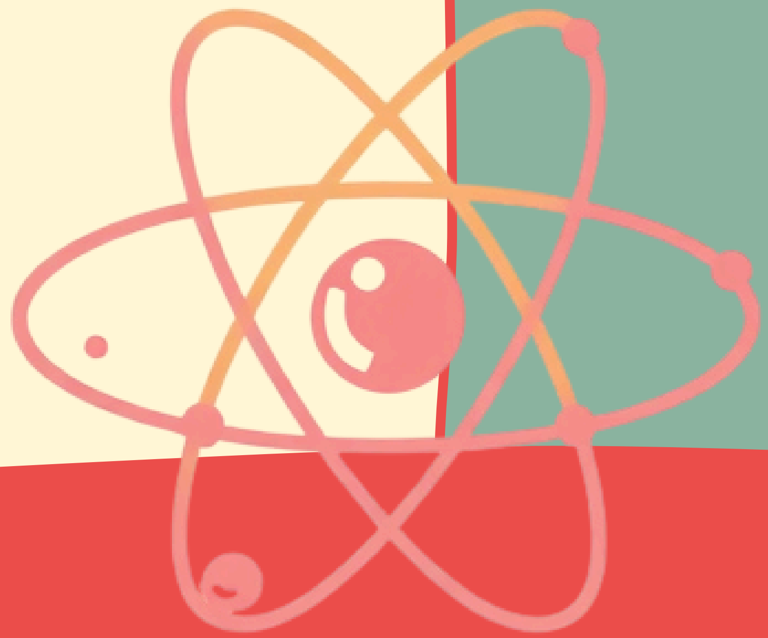
Fusion energy has amazing potential! It naturally powers the Sun and all the stars in our universe. Scientists around the world are working hard to harness fusion energy here on Earth for clean electricity. The best part? Fusion produces much less radioactive waste than fission, making it safer for our planet. Imagine a future with unlimited clean energy!

Fissión vs Fusion

Nuclear Fission

- Splits a big nucleus into smaller parts
- Used in current nuclear power plants
- Produces radioactive waste that needs careful handling
- Powers electricity, submarines, and medical treatments





VS

Nuclear Fusion

- Joins small nuclei to make a bigger one
- Powers the Sun and all stars naturally
- Cleaner and safer with less waste
- Still experimental – scientists working on it!

Safety & Environmental Impact

Nuclear power plants have many safety systems to protect people and the environment. Radioactive waste from fission must be stored and handled very carefully for thousands of years. Scientists use thick concrete walls and special containers to keep everyone safe. Fusion energy is much cleaner and safer because it produces very little radioactive waste. That's why researchers are excited about fusion as a future energy source!

The Future of Nuclear Energy

Scientists are working hard to make fusion energy practical for everyday use. Nuclear energy can help reduce pollution and fight climate change, making our planet cleaner and healthier. The most exciting part? You could be the scientists who discover new ways to harness energy! Your curiosity today could power tomorrow's world.

Recap: What We Learned

Energy is the power to do work and it powers everything around us! Nuclear energy comes from the center of atoms called the nucleus. It's one of the most powerful energy sources we have.

Fission splits big nuclei into smaller parts, releasing energy. Fusion joins small nuclei together, releasing even more energy! Both are used in different ways, and fusion could be the future of clean energy.

Thank You!

Let's Discuss